

Mapping Invasive Alien Plant Species Across Virginia, United States — Project Workflow

Raymond Santiago and Seun Oladipo.

Data/References

We have boundary data for Virginia and the United States. For the Land Use Land Cover datasets, we are still determining the most efficient way to incorporate them into R. We are currently considering loading the data through the ESRI API.

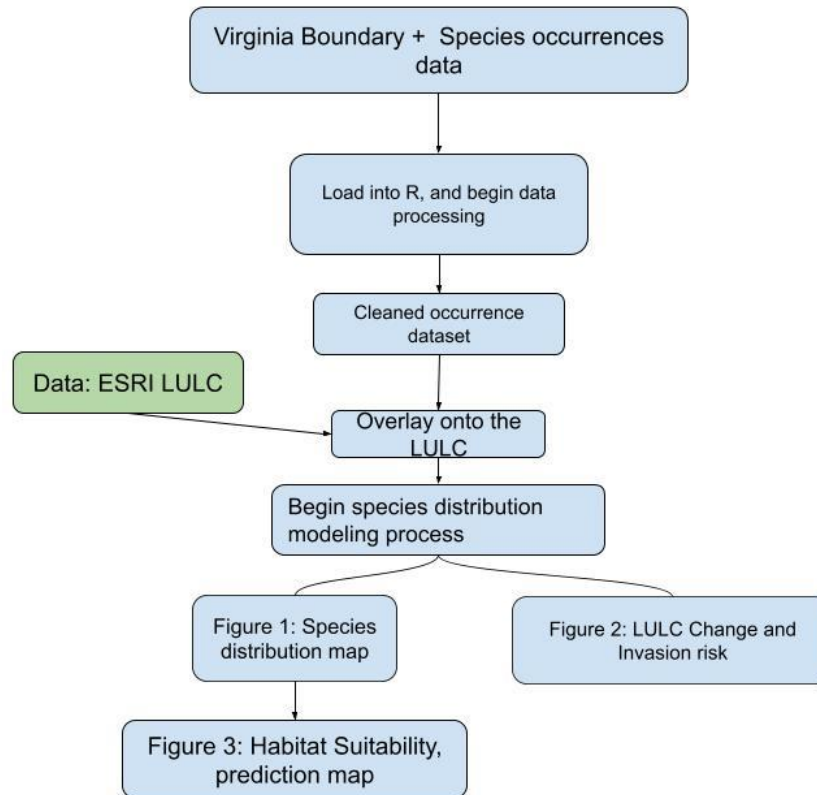
Finding the species distribution dataset has proven difficult; we are unable to find a specific dataset for invasive plant species in Virginia and are still searching. We've explored possible sources such as iNaturalist, Global Biodiversity Information Facility (GBIF), and the Virginia Department of Conservation and Recreation. For example, GBIF contains plant species across Virginia. When filtering the data for invasive species, the search returned only one plant species, which does not meet our analysis criteria. This highlights the primary step of our workflow: full acquisition of the dataset that aligns with our research question.

We are also considering pivoting our research question to examine endangered, vulnerable, or extinct plant species. This alternative would still allow us to examine spatial patterns in species distributions and their relationships with land use and environmental variables. Although not all the datasets have been fully obtained, we have identified potential sources:

Data Name	Source	Description	Temporal Extent and Resolution	Spatial Extent and Resolution	Data Found
ESRI Land Use Land Cover	ESRI	Land use change	2017-2023	2m resolution	Yes
Virginia Boundary	ESRI Living Atlas	Study area boundary	--	--	yes
Temperature	NOAA	--	1851- Present	Global	No
Precipitation	NOAA	--	1851- Present	Global	No
Species Distribution Data	Virginia Department of Conservation and Recreation	--	--	Virginia	No

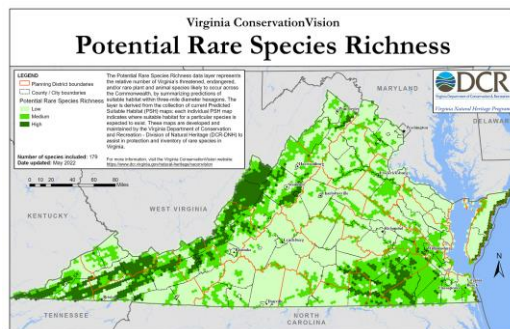
Workflow Diagram

Species Distribution Map Workflow



Proposed Figures

Potential figure 1: Species distribution map



Source: Virginia Department of Conservation and Recreation, Division of Natural Heritage. (2022). *Potential rare species richness* [Map]. Virginia ConservationVision. <https://www.dcr.virginia.gov/natural-heritage/vaconvisprsr>

- Figure 1 will be a map of Virginia showing the locations of invasive plant species overlaid on the LULC classification layer; this will most likely be produced using ggplot2.
- Figure 2 will present temperature and precipitation maps of Virginia, showing the climate context of where species occurrences are concentrated and how environmental conditions support and limit the presence of invasive species.
- Figure 3, our most complex figure, aims to show how land may transition and whether this is associated with the presence of invasive species. This will combine the temporal LULC data with the occurrence records to hopefully identify which types of land disturbances most predict invasion.
 - o Land-use types are critical factors in species' habitats. Therefore, our result may also illustrate the land use map in the continental USA and the land use map of Virginia.

Revised Research Questions and Likely Gaps in Knowledge

Our research question remains the same: a spatial analysis of invasive plant species in Virginia. We're majorly doing a spatial analysis of invasive plant species distribution in Virginia using R. We will pivot to examining endangered or vulnerable plant species using the same framework. This analysis could be done using GIS software such as QGIS and ArcGIS, but this project will enable us to explore the capabilities of the R programming language for spatial analysis.

Workload distribution

The conceptualization, writing, data handling, and coding will be equally done. Code sharing will either be done using a shared OneDrive, or if there reaches the need, a GitHub repository may be utilized.